Building Queries: An Exploration

```
$ echo "Data Sciences Institute"
```

Building Queries:

→ Fundamental Three Commands

Two More Commands

Putting Things Together with JOIN

Fundamental Three Commands

Fundamental Three Commands

- Our first three commands (`SELECT` , `FROM` , `WHERE`) are essential to nearly every SQL query
- The template for our initial SQL statement is as such:

  `SELECT` : *the columns we want to retrieve*

  `FROM` : *the table we are querying*

  `WHERE` : *filters/conditions (optional)*

  `ORDER BY` : *column sorting: ascending* `ASC` *or descending* `DESC` *(optional)*

  `LIMIT` : *how many rows we want to return (optional)*

Fundamental Three Commands

- Always specified in this order:
  - `SELECT` will come first
  - `FROM` will come after `SELECT`
    - when we are querying more than one table at a time, each will come after `FROM` but before `WHERE` (more on this later)
  - `WHERE` will come after `FROM`
  - `ORDER BY` will come after `WHERE` clauses

Fundamental Three Commands

- We'll sometimes use the `LIMIT` clause to look at data
  - This comes at the very end of a query
  - `LIMIT` shouldn't be used for analytics unless you have a specific reason
    - `ORDER BY` often impacts the usefulness of `LIMIT`
- Remember:
  - In SQL, we use two dashes `--` to comment out lines, rather than `#`

SELECT Command

- At its simplest `SELECT` specifies column names we are retrieving
    - commas come between each column name
        - `SELECT student, course, grade ...`
    - column names with a space need to be enclosed in square brackets
        - `SELECT [poorly named column], better_column_name, AnotherColumnName`

SELECT Command

- Within `SELECT` statements we can perform manipulations on columns
  - e.g. rename a column
    - `SELECT [poorly named column] AS better_col`
  - combine two text columns
  - perform math on a numeric column
  - ...and many more things

SELECT Command

- We can use `SELECT` to perform math without a `FROM` statement
  - `SELECT 1 + 1`
  - `SELECT 10*5, cos(2), pi()`
- And we can use `SELECT` to specify constant values
  - `SELECT 2026 AS this_year, 'July' AS this_month`
- When selecting columns, they need to exist in the table!

FROM Command

- `FROM` statements indicate which table the data is from and where the table is located
    - in more complicated RDBMs, you will often have multiple databases on the same server and multiple schema within those databases
        - a fully qualified location of a table would thus be `database.schema.table`
- `SELECT * FROM table_name` indicates *everything* in the table
- Best practice suggests that we should explicitly call each column, even if we want all of them
    - **Why do we think this is the case? 💬💭 Think, Pair, Share**

SELECT & FROM

(`SELECT` & `FROM` live coding)

WHERE Command

- `WHERE` clauses are conditions that the query will follow
- When we want to have multiple conditions, we use a single `WHERE` and then additional logical operations
- `WHERE` clauses always return rows evaluating to `TRUE`
    - Follows Boolean rules if more than one condition is present

WHERE Command

```
SELECT *
FROM students
WHERE first_name = 'Thomas'
AND last_name = 'Rosenthal'
```

- **Notice we put string values in single quotes**
    - SQLite also allows double quotes, with a few minor caveats

WHERE Command

Logical Operators

- `AND`
- `OR`
- `NOT`
- `NOT IN`
- equals: `=`
- does not equal: `<>` `!=`
  - (flavour dependent)

WHERE Command

Logical Operators (continued...)

- greater than (equal to): `>` `>=`
- less than (equal to): `<` `<=`
- `BETWEEN`
- `EXISTS`
  - table specific
- `IS`
  - `NULL` specific

WHERE Command

- `NULL` is not a value (it's the absence of a value)
  - to check null values, we use `IS NULL` or `IS NOT NULL`
  - `= NULL` will not work

WHERE Command

- `LIKE` allows for string wildcards
- `%` specifies the wildcard placement
  - `country_name LIKE 'and%'`
    - Andorra
  - `country_name LIKE '%and'`
    - Finland, Iceland ...more
  - `country_name LIKE '%and%'`
    - all of the above, *plus* Antigua and Barbuda, Netherlands, Rwanda ...more!
  - `country_name LIKE '%an%d%'`
    - Canada ...surely more!

WHERE Command

(`WHERE` live coding)

What questions do you have about `SELECT`, `FROM`, `WHERE`?

Building Queries:

Fundamental Three Commands

→ Two More Commands

Putting Things Together with JOIN

Two More Commands

- `CASE` : Implements conditional logic.
- `DISTINCT` : Returns unique values.

CASE Command

- `CASE` statements allow us to introduce conditional logic into our `SELECT` statements

CASE Command

- They are generally similar to `if` or `if else` statements in python, R, and other languages
  - When a condition is introduced, we check whether it evaluates to TRUE
    - If it is true, we proceed with a desired command, calculation, value, etc
    - If it is not true, we move to the next condition
      - If it is true, we proceed with another desired command, calculation, value, etc
        - ...all the way until we run out of conditions
  - For all FALSE conditions, we can use an `ELSE` statement if we want to

CASE Command

- The results of a `CASE` statement will be a new column
- Best practice is to name the new column using `AS new_column_name`

```
CASE
  WHEN [something is true]
    THEN [value or calculation]
  WHEN [something else is true]
    THEN [value or calcuation]
  ELSE [value or calcuation]
END
```

CASE Command

(`CASE` live coding)

DISTINCT Command

- Not all queries will result in unique rows (i.e. duplicates are present)
    - **Can we think of why this is? Write your thoughts in the etherpad!**

DISTINCT Command

- `DISTINCT` has two possible spots within a query:
  - One comes immediately after `SELECT`, before column names are specified
    - e.g. `SELECT DISTINCT songs, albums, artists...`
    - This `DISTINCT` will govern the entire query
  - The other comes within aggregation (we'll get to this later)
    - e.g. `COUNT(DISTINCT products)`
    - This `DISTINCT` will only affect this specific aggregation

DISTINCT Command

(`DISTINCT` live coding)

Building Queries:

Fundamental Three Commands

Two More Commands

→ Putting Things Together with JOIN

Joining Tables

- Joins are used to combine data stored in different tables into a single table

Joining Tables

- Joins are the "Cartesian product" of two tables with *conditional selection(s)* of specific rows
  - A Cartesian product combines all possible row values with another
    - An easy example is a deck of cards:

      combining four suits:
      {♠, ♥, ♦, ♣}

      with thirteen ranks:
      {A, K, Q, J, 10, 9, 8, 7, 6, 5, 4, 3, 2}

      produces 52 cards (4 * 13)
  - To create a Cartesian Product in SQL we use `CROSS JOIN` (rare, but not unheard of)

Joining Tables

- Joins require relationships (with one exception, `CROSS JOIN`) between tables
- Different joins create different results
  - Join names specify which conditional selection is desired

Joining Tables

- There are three join types in SQL but different joining criteria can further limit results
- The most permitting join is a `FULL OUTER JOIN` and the least permitting is an `INNER JOIN`
  - Let's explore what this means by looking at each of them

JOIN Syntax

Syntax for a join is as follows:

```
SELECT [columns]

FROM [left table]

JOIN [right table]

ON [left table.matching column] = [right table.matching column]
```

Joining Tables

A couple of notes:

- You will need to specify which join type is desired:
  - e.g. `INNER JOIN`
- Matching columns do not need to have the same name, just the same value
  - e.g. `ON table1.LetterGrade = table2.Alphabet` will work because A=A, B=B, C=C, etc
- You can specify more than one column to be joined
  - e.g. `ON table1.FirstName = table2.FirstName AND table1.LastName = table2.LastName`

INNER JOIN

- `INNER JOIN` filters both tables to rows present in both tables
- `INNER JOIN` does not produce `NULL` values
- `INNER JOIN` is the "default" join
  - i.e. queries do not need to specify "INNER", though it's good practice to write INNER

Source: Image: Teate, Chapter 5 (+annotations by me)

Inner Join

Only rows from the "right table" and "left table" where values in the specified fields have matches in both tables

Colour	Quantity
Pink	1
Blue	1
Green	2
Yellow	2
Black	6
Orange	3
Red	1
Purple	1

Colour	Quantity
Pink	1
Teal	3
Green	2
Gold	2
Black	6
Orange	3
Crimson	2
Purple	2

Inner Join Result Set

	Colour	Quantity	Colour	Quantity
1	Pink	1	Pink	1
2	Green	2	Green	2
3	Black	6	Black	6
4	Orange	3	Orange	3

INNER JOIN

A quick note on table aliasing:

- It is very common practice to alias table names
    - It makes join criteria much more concise
    - It simplifies `SELECT` statements when column names are the same
        - This is a common error: *"ambiguous column name"*
            - SQL requires you to specify *which* table you are returning the result from
- Generally, tables are aliased with the first letter (or first few letters) of the table so they can be easily referenced
    - `product AS p`
    - `product_category AS pc`

INNER JOIN

(`INNER JOIN` live coding)

LEFT (OUTER) JOIN

- `LEFT JOIN` filters the "right" table to rows present in the "left" table
- `LEFT JOIN` will most often produce `NULL` values
- The "OUTER" in `LEFT OUTER JOIN` is optional
  - Generally, OUTER seems to be excluded, but both are correct
- LEFT *is not* optional; there is no "OUTER JOIN"

Source: Image: Teate, Chapter 5 (+annotations by me)

Left Join

All rows from the "left table", and only rows from the "right table" with matching values in the specified fields

Colour	Quantity
Pink	1
Blue	1
Green	2
Yellow	2
Black	6
Orange	3
Red	1
Purple	1

Colour	Quantity
Pink	1
Teal	3
Green	2
Gold	2
Black	6
Orange	3
Crimson	2
Purple	2

Left Join Result Set

	Colour	Quantity	Colour	Quantity
1	Pink	1	Pink	1
2	Blue	1	NULL	NULL
3	Green	2	Green	2
4	Yellow	2	NULL	NULL
5	Black	6	Black	6
6	Orange	3	Orange	3
7	Red	1	NULL	NULL
8	Purple	1	NULL	NULL

LEFT (OUTER) JOIN

(`LEFT JOIN` live coding)

RIGHT (OUTER) JOIN

- `RIGHT JOIN` filters the "left" table to rows present in the "right" table
- `RIGHT JOIN` will most often produce `NULL` values
- The "OUTER" in `RIGHT OUTER JOIN` is optional
  - Generally, OUTER seems to be excluded, but both are correct

Source: Image: Teate, Chapter 5 (+annotations by me)

Right Join

All rows from the "right table", and only rows from the "left table" with matching values in the specified fields

Colour	Quantity
Pink	1
Blue	1
Green	2
Yellow	2
Black	6
Orange	3
Red	1
Purple	1

Colour	Quantity
Pink	1
Teal	3
Green	2
Gold	2
Black	6
Orange	3
Crimson	2
Purple	2

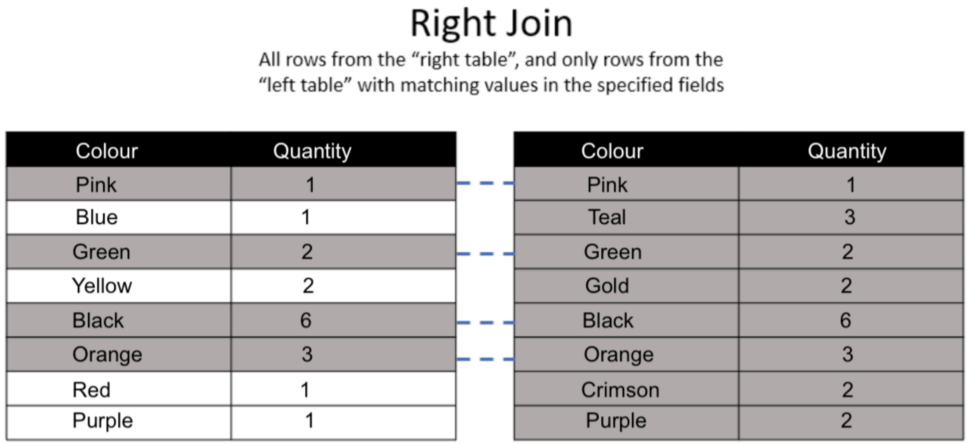
Right Join Result Set

	Colour	Quantity	Colour	Quantity
1	Pink	1	Pink	1
2	Green	2	Green	2
3	Black	6	Black	6
4	Orange	3	Orange	3
5	NULL	NULL	Teal	3
6	NULL	NULL	Gold	2
7	NULL	NULL	Crimson	2
8	NULL	NULL	Purple	2

RIGHT (OUTER) JOIN

- `RIGHT JOIN` is somewhat frowned upon, but sometimes they make sense
  - Often your query can be reorganized to use a `LEFT JOIN` instead

Source: Image: Teate, Chapter 5



Colour	Quantity
Pink	1
Blue	1
Green	2
Yellow	2
Black	6
Orange	3
Red	1
Purple	1

Colour	Quantity
Pink	1
Teal	3
Green	2
Gold	2
Black	6
Orange	3
Crimson	2
Purple	2

FULL (OUTER) JOIN

- `FULL OUTER JOIN` does not filter either "left" or "right" table
- Expect `NULL` values to be produced from a `FULL OUTER JOIN`
- My experience has been to write `FULL OUTER JOIN` rather than `FULL JOIN` but this is personal preference

Full Outer Join Result Set

	Colour	Quantity	Colour	Quantity
1	Pink	1	Pink	1
2	Blue	1	NULL	NULL
3	Green	2	Green	2
4	Yellow	2	NULL	NULL
5	Black	6	Black	6
6	Orange	3	Orange	3
7	Red	1	NULL	NULL
8	Purple	1	NULL	NULL
9	NULL	NULL	Teal	3
10	NULL	NULL	Gold	2
11	NULL	NULL	Crimson	2
12	NULL	NULL	Purple	2

Filtering a FULL (OUTER) JOIN

- All OUTER JOIN syntax can be filtered to exclude the *matching* criteria
  - Often called an ANTI JOIN, i.e. what's *not* in the other table

```
SELECT *
FROM table_1
{LEFT | RIGHT | FULL} OUTER JOIN table_2
ON table_1.key = table_2.key
WHERE {table_1.key IS NULL | table_2.key IS NULL |
       table_1.key IS NULL OR table_2.key IS NULL}
```

Anti Join Result Set

	Colour	Quantity	Colour	Quantity
1	Blue	1	NULL	NULL
2	Yellow	2	NULL	NULL
3	Red	1	NULL	NULL
4	Purple	1	NULL	NULL
5	NULL	NULL	Teal	3
6	NULL	NULL	Gold	2
7	NULL	NULL	Crimson	2
8	NULL	NULL	Purple	2

Multiple Table Joins

- More than one table can be joined at a time

```
SELECT *
FROM table_1
{INNER | LEFT | FULL JOIN} table_2
  ON table_1.key = table_2.key
{INNER | LEFT | FULL JOIN} table_3
  ON {table_1 | table_2}.key = table_3.key
{INNER | LEFT | FULL JOIN} table_n
  ON {table_1 | table_2 | table_3}.key = table_n.key
```

Multiple Table Joins

- The order and type of joins will have significant effect on the final result set
- It's important to determine which table should be the `FROM` table
- Sometimes you have to experiment a bit to get things right
- **Can you imagine scenarios based on your knowledge of different `JOIN` types that result in significantly different outputs?**

Multiple Table Joins

(Multiple Table Joins live coding)

What questions do you have about anything from today?